

UCS531: CLOUD COMPUTING

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Course Objectives: To learn the concepts of cloud infrastructure and services in addition with its implementation for assessment of understanding the course by the students.

Introduction and Evolution of Computing Paradigms: General Benefits and Architecture, Business Drivers, Main players in the Field, Overview of Existing Hosting Platforms and its architecture, Cluster Computing, Grid Computing, XaaS Cloud Based Service Offerings, Overview of Security Issues

Classification of Cloud Implementations: Key Amazon offerings-Amazon Web Services, The Elastic Compute Cloud (EC2), Simple Storage Service (S3), Simple Queuing Services (SQS), Bundling Amazon instances, AWS Identity Management and Security in the Cloud, Messaging in the Cloud, RESTful Web Services.

Virtualization: Virtualization, Advantages and disadvantages of Virtualization, Types of Virtualization: Resource Virtualization i.e. Server, Storage and Network virtualization, Migration of processes, Classic Data Center, Virtualized Data Center (Compute, Storage, Networking and Application), Business Continuity in VDC. VMware vCloud – IaaS, Network virtualization through Software Defined Networks

Cloud based Data Storage: Introduction to Hadoop, Hadoop Ecosystem (Pig, Hive, Cassandra and Spark), Introduction No-SQL databases, Map- Reduce framework for Simplified data processing on Large clusters using Hadoop, Data Replication, Shared access to data stores.

Related Technologies: Introduction to Fog Computing and Edge Computing, Usage of Cloud for IoT and Big data analytics, Overview of Google AppEngine - PaaS, Windows Azure

Self-learning Content:

Cloud Issues and Challenges: Cloud models, Cloud computing issues and challenges like Security, Elasticity, Resource management and Scheduling, QoS (Quality of Service) and Resource Allocation, Cost Management and Cloud bursting.

Laboratory work:

To implement Cloud, Apache and basics of Hadoop framework, an open source implementation of MapReduce, and its Java API, Hadoop Distributed File System (HDFS). Implementation of RESTful Web Services. To understand various concepts about virtualization and data storage. To implement few algorithms with the help of MapReduce and some high-level language.

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Comprehend the basic concepts and architecture of Cloud computing.
2. Implement Cloud Services through AWS offerings and Restful web services.
3. Apply the knowledge of virtualization through different virtualization technologies.
4. Perform operations on data sets using Map Reduce framework, SQL and NO SQL databases.

Text Books:

1. Buyya K, R., Broberg J. and Goscinski M. A., Cloud Computing: Principles and paradigms, MIT Press (2011) 4th ed.
2. Kai Hwang, Geoffrey Fox and Jack Dongarra, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Morgan Kaufmann (2012) 2nd ed.
3. Miller M., Cloud Computing, Que Publishing (2008) 1st ed.
4. Puttini R. and Mahmood Z., Cloud Computing: Concepts, Technology & Architecture, Service Tech press (2013) 1st ed.

Reference Books:

1. Velte A., Velte T., and Elsenpeter R., Cloud Computing: A practical Approach, Tata McGrawHill (2009) 1st ed.
2. Hurwitz J., Bllor R., Kaufman M. and
3. Halper F., Cloud Computing for dummies (2009) 1st ed.

Evaluation Scheme

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30